Practical – 7: Process Address Space and Memory Segments

Aim

To write a C program that prints the addresses of code, global, static, heap, and stack variables and analyze the Linux process address space using /proc/PID/maps.

```c
#include <stdio.h>
#include <stdlib.h>

int global = 100;
static int s = 50;

int main()
{
    int stack = 10;
    int *heap = (int *)malloc(sizeof(int));

    printf("Code Segment  : %p\n", (void *)main);
    printf("Global Segment: %p\n", (void *)&global);
    printf("Static Segment: %p\n", (void *)&s);
    printf("Heap Segment  : %p\n", (void *)heap);
    printf("Stack Segment : %p\n", (void *)&stack);

    free(heap);

    return 0;
}
```

```
meghana@hp1: ~
meghana@hp1:~$ nano practical7.c
meghana@hp1:~$ gcc practical7.c -o practical7     actical7
meghana@hp1:~$ ./practical7
Code Segment  : 0x647ab801b1a9
Global Segment: 0x647ab801e010
Static Segment: 0x647ab801e014
Heap Segment  : 0x647af40f3010
Stack Segment : 0x7ffd62aa30ec
meghana@hp1:~$
```